

Sidharth Vijayan Krishnan

Software Developer

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github.com/Sidharthkris | linkedin.com/in/sidharth-vijayan-krishnan | Authorized to work in Germany

SUMMARY

M.Sc. Computer Science graduate with hands-on software development experience across the full SDLC, specializing in modern full-stack web applications. Based in Hannover and available immediately. Proficient in English (C2) with professional working proficiency in German (B1, actively advancing to B2).

SKILLS

- Programming Languages** — Java, Python, JavaScript, TypeScript, C++, SQL
- Backend Frameworks & Technologies** — Spring Boot, Spring Security, Spring Data JPA, Thymeleaf, REST APIs
- Frontend Development** — React, Angular, HTML5, CSS3, Vite, React Router
- Databases** — PostgreSQL, MySQL, Microsoft SQL Server
- DevOps & Build Tools** — Git, Docker, Linux, CI/CD, GitHub Actions, Jenkins, Maven
- Data Science & Analytics** — Pandas, NumPy, Matplotlib, Jupyter
- Software Testing & QA** — Selenium WebDriver, Appium, JUnit, TestNG, Vitest, JMeter
- Methodologies & Practices** — Agile, Scrum, TDD, BDD

EDUCATION

M.Sc. Computer Science <i>Technische Universität Clausthal</i>	01.2021 – 09.2025 <i>Clausthal, Germany</i>
B.Tech. Computer Science and Engineering <i>Mar Baselios Institute of Technology and Science (MBITS)</i>	08.2011 – 05.2015 <i>Kerala, India</i>

EXPERIENCE

Test Automation Engineer (Internship) <i>Rogersoft Technologies Pvt. Ltd.</i>	12.2018 – 03.2019 <i>Kerala, India</i>
- Built automated web and mobile testing frameworks across CI/CD pipelines. - Automated 150+ web regression tests via Selenium and TestNG, cutting cycle time by 40%. - Implemented mobile test automation with Appium, achieving 85% coverage on iOS and Android. - Integrated automated test suites into Jenkins, GitLab CI, and GitHub Actions (feedback time under 2 hours). - Executed load and stress tests using JMeter for 5,000+ virtual users, fixing 3 critical bottlenecks. - Standardized build configurations and dependencies using Maven, reducing setup time by 30%.	
Software Developer & Academic Coordinator <i>Matrix Engineering Pvt. Ltd.</i>	11.2015 – 11.2018 <i>Kerala, India</i>
- Engineered full-lifecycle Java applications (JDBC, MySQL) automating grading, attendance, and student record management. - Built backend business logic and database procedures to streamline result generation and performance tracking. - Delivered training and project mentoring in Core Java, OOP, Data Structures, Algorithms, SQL, and Servlets/JSP. - Created technical documentation, coding assignments, and standardized evaluation frameworks. - Coordinated academic workflows, scheduling, and curriculum operations across five departments.	
Full-Stack Developer (Internship) <i>Verbicio Labs Pvt. Ltd.</i>	07.2015 – 10.2015 <i>Kerala, India</i>
- Built responsive UI components using HTML5, CSS3, and JavaScript, boosting user engagement by 15%. - Designed and optimized relational SQL database schemas for 10,000+ user records. - Resolved 30+ full-stack bugs to ensure release readiness and production stability.	

RESEARCH & PROJECTS

Agent-Based Simulation of Emergency Evacuation in University Lecture Halls

09.2024 – 09.2025

Master Thesis (Agent-Based Modelling, Multi-agent System, Netlogo, Python)

- Developed an agent-based model in NetLogo to evaluate emergency evacuation dynamics across 5 lecture hall layouts.
- Simulated 200+ heterogeneous agents incorporating behavioral psychology, mobility constraints, and decision-making rules.
- Automated 9,000+ experiments via NetLogo BehaviorSpace to sweep exit parameters and door blockage scenarios.
- Engineered Python data pipelines to analyze and visualize evacuation times, flow rates, and bottleneck formation.

Agent-Oriented Programming: Designing BDI Agents in GOAL for an Exploration and Coordination Problem Based on the MAPC

03.2024 – 08.2024

Master Project (Agent-Oriented Programming, Multi-Agent System, GOAL, Java)

- Programmed autonomous BDI cognitive agents in GOAL and Java for real-time hex-grid exploration.
- Implemented decentralized coordination and task-allocation algorithms for multi-agent teams without a central controller.
- Designed agent communication protocols to share spatial beliefs and optimize collective mapping efficiency.
- Formulated logic-based decision rules for dynamic pathfinding and obstacle avoidance under time constraints.

Course Timetable Planner (Full-Stack SPA & REST API) [↗](#)

React 19, TypeScript, Java 17, Spring Boot 3, PostgreSQL, Docker, Vitest, JUnit 5

- Developed a decoupled course-scheduling system using Java 17, Spring Boot 3, and PostgreSQL with automated conflict detection.
- Built a single-page frontend using React 19, TypeScript, and Vite with a typed API integration layer.
- Implemented role-based access control (RBAC) via Spring Security across all REST endpoints and client views.
- Containerized the stack with Docker, documented APIs using OpenAPI/Swagger, and wrote unit/integration test suites with JUnit 5, Vitest, and React Testing Library.

Quiz Assessment Engine [↗](#)

Java 17, Swing, Jackson, JUnit 5, Maven

- Built an event-driven Java 17 assessment engine using GoF patterns (Factory, Builder, Strategy, Observer) to decouple core modules.
- Developed a Java Swing GUI featuring live timers, dynamic question randomization (120+ items), and session leaderboards.
- Implemented polymorphic JSON data persistence and serialization using Jackson.
- Achieved 100% test coverage with JUnit 5 and Mockito, including headless GUI testing for Swing components.

LANGUAGES

English (C2, Fluent)

German (B1, Actively working toward B2)

Malayalam (Native)

CERTIFICATES

Ethical Hacker (Cisco)

Introduction to Cybersecurity (Cisco)